

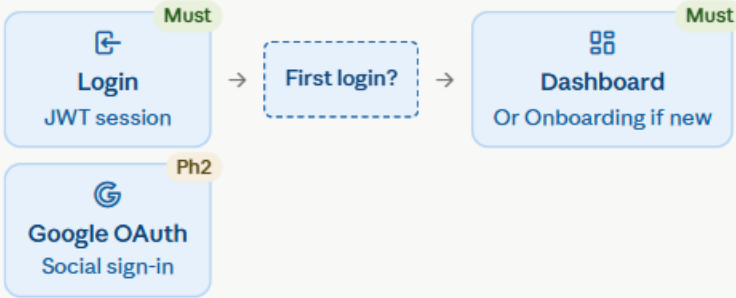
🌐 1 — Entry point

Public → authenticated access

NEW USER REGISTRATION PATH



RETURNING USER PATHS — FIX 1: LOGIN NOW ARROWS TO DASHBOARD



FIX 2: FORGOT PASSWORD NOW LINKED TO OTP SYSTEM



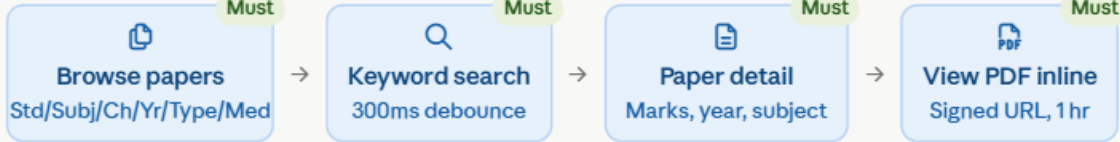
🎓 2 — Student portal

Protected — JWT required

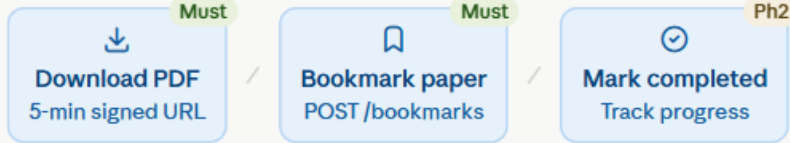
DASHBOARD WIDGETS



FIX 3: PAPERS BROWSE — ALL 6 FILTERS NOW SHOWN



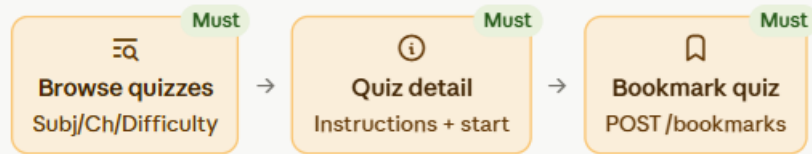
ACTIONS ON A PAPER



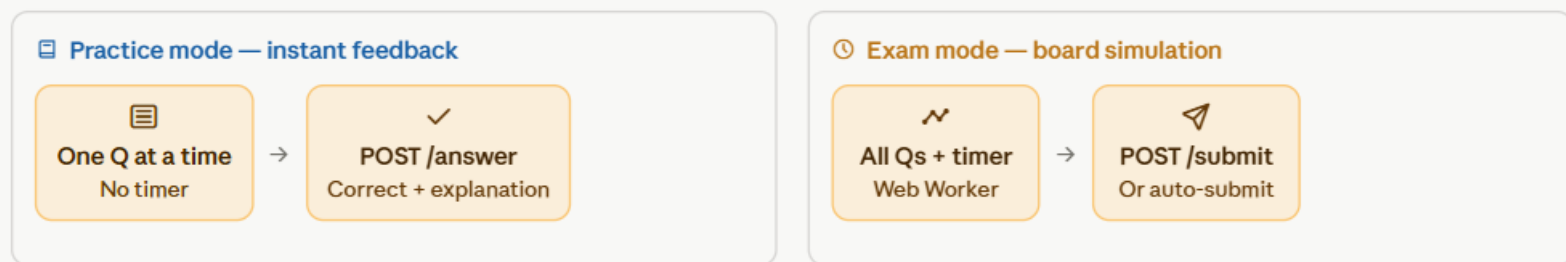
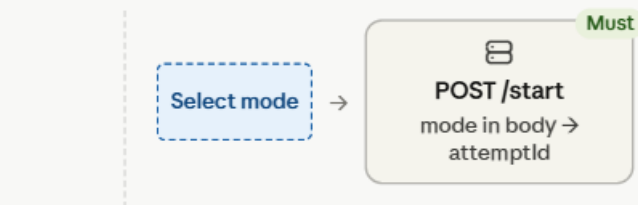
BOOKMARKS PAGE + PROFILE



FIX 6: BOOKMARK QUIZ ADDED ON DETAIL PAGE

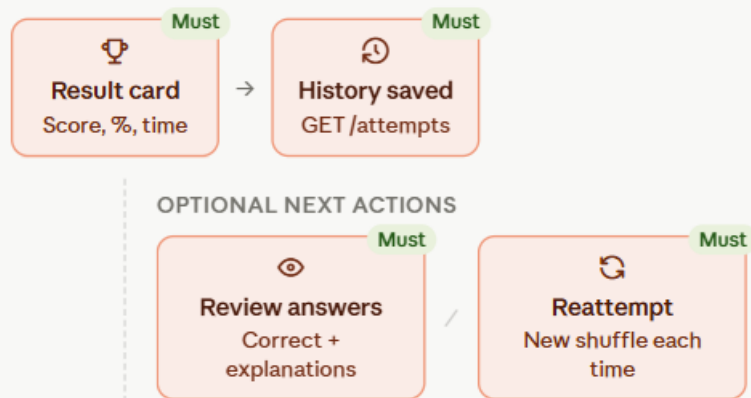


FIX 4: MODE SELECTED BEFORE POST /START — MODE IS SENT IN REQUEST BODY



↓

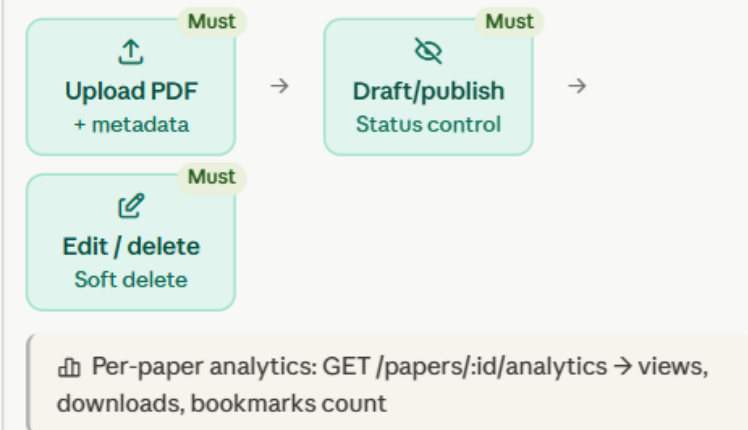
FIX 5: HISTORY SAVED IMMEDIATELY AFTER SUBMIT — REVIEW + REATTEMPT ARE SEPARATE OPTIONAL ACTIONS



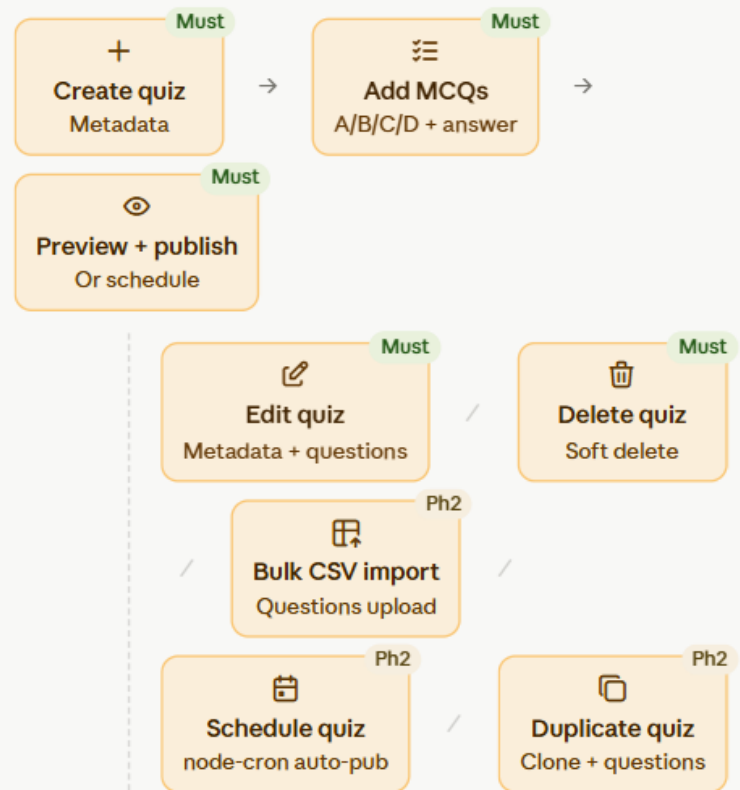
ADMIN AUTH



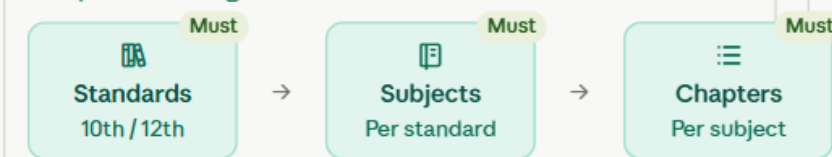
Paper management



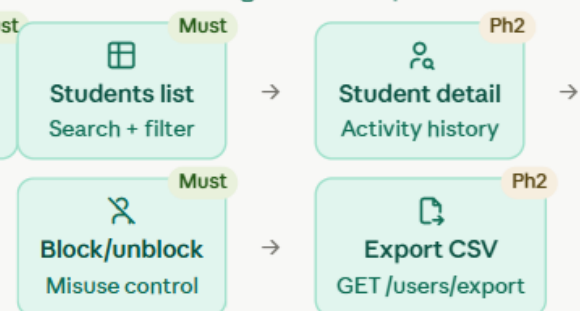
FIX 7: Quiz management — edit + delete added



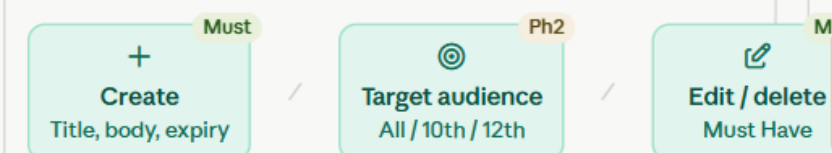
Syllabus management



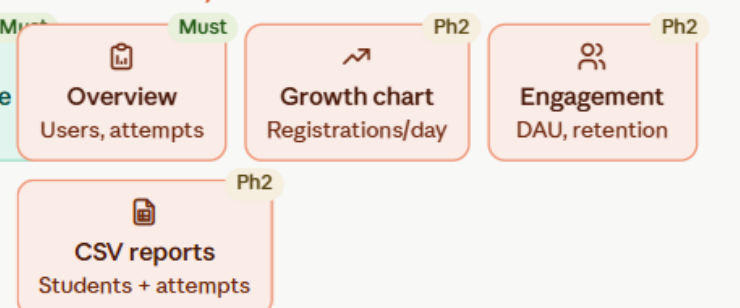
FIX 8: User management — Export CSV added



Announcements



Platform analytics



5 — Backend systems

Express.js API + supporting infrastructure



JWT auth

15 min / 7 day refresh



Cloudinary

PDFs + signed URLs



Redis / Upstash

5–60 min TTL



SendGrid

OTP +
announcements



node-cron

Scheduled auto-pub



Sentry

Error tracking



MongoDB Atlas

M10, Mumbai region



6 — Auth / system layer

Shared infrastructure — runs behind every authenticated ses ...

JWT TOKEN LIFECYCLE



Login success

Credentials verified

Must



Issue tokens

Access 15 min +
refresh 7d

Must



API request

Bearer header

Must



Refresh on 401

Silent renewal

Must



Logout

Invalidate all tokens

Must

① Refresh token rotates on every use · Up to 5 active refresh tokens stored per user (multi-device) · Admin session auto-expires after 30 min inactivity

OTP VERIFICATION FLOW — USED BY BOTH REGISTER AND FORGOT PASSWORD



Trigger OTP

Register / forgot pwd

Must



Generate code

6-digit, 10-min TTL

Must



Email via SendGrid

Dynamic template

Must



User enters OTP

6-digit input

Must



Verify + act

bcrypt compare

Must

① Rate limit: max 3 OTP emails/hr per address · Redis key: `email_limit:{email}` TTL 1 hr · Max 5 failed attempts before regeneration required

RBAC MIDDLEWARE CHAIN — EVERY PROTECTED ROUTE



authenticate()

Verify JWT, attach
user

Must



authorize(role)

student / admin

Must



verifyAdminKey()

X-Admin-Key header

Must



Route handler

Controller → Service

Must

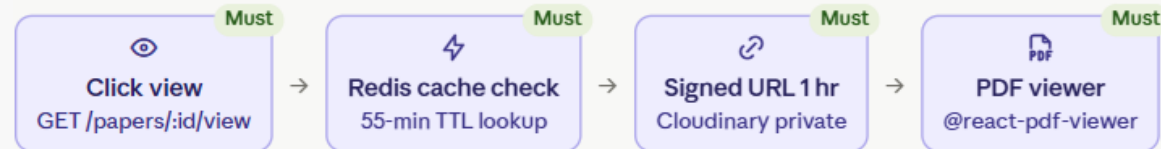
7 — File storage pipeline

Cloudinary CDN — PDF upload, private signed URLs, media delivery

ADMIN PDF UPLOAD

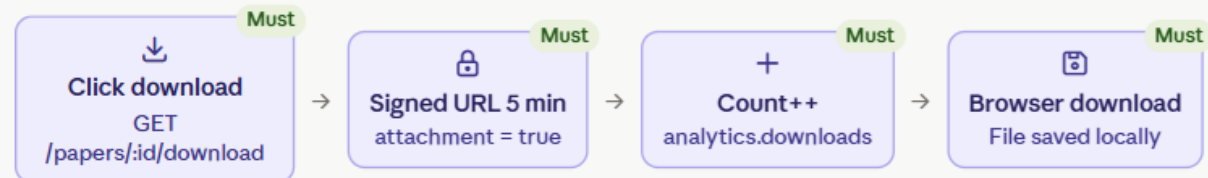


STUDENT — VIEW PDF INLINE

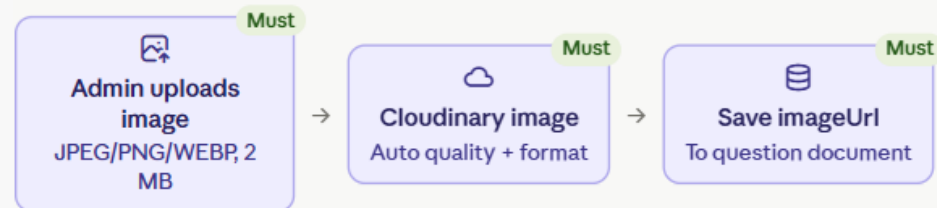


① pdfUrl is never exposed in API responses — only a server-generated signed URL is returned, expiring after 1 hr · Cached in Redis to avoid regenerating on every page load

STUDENT — DOWNLOAD PDF



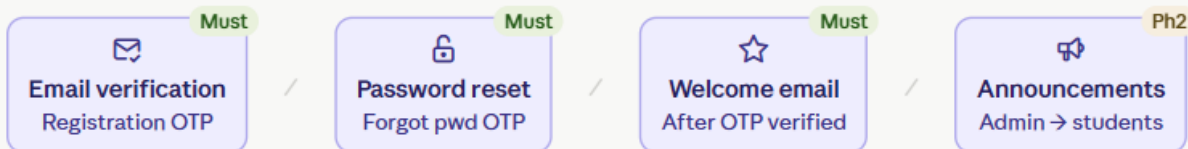
QUESTION IMAGE UPLOAD (FOR MCQ QUESTIONS WITH IMAGES)



8 — Email & notifications

SendGrid transactional emails — OTPs, welcome, announcements

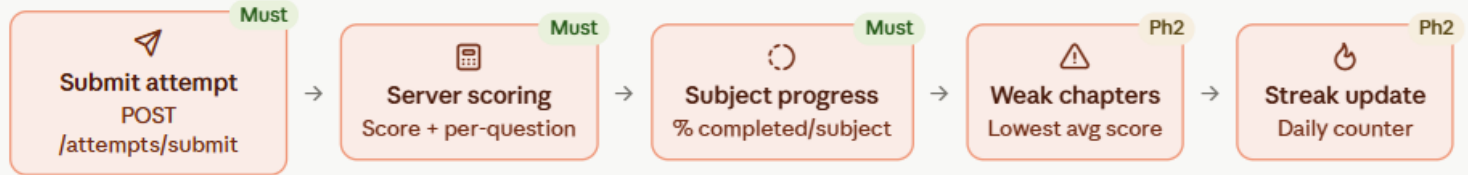
EMAIL TRIGGERS AND TEMPLATES



① Rate limit: max 3 OTP emails/hr per address · Redis key: email_limit:{email} TTL 1 hr · Returns 429 if exceeded

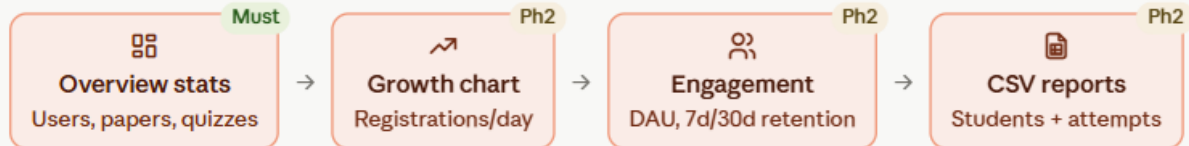
</> SendGrid dynamic templates · dynamicTemplateData: { otp, expiresIn: '10 minutes', year } · From: noreply@mahaboardprep.in

STUDENT ANALYTICS PIPELINE



ⓘ Dashboard data cached in Redis for 5 min per user · Subject progress = (quizzes attempted in subject / total quizzes in subject) × 100
· Streak resets if no activity for 24 hrs

ADMIN ANALYTICS PIPELINE



ⓘ Admin overview stats cached in Redis for 10 min · Per-question correct % identifies ambiguous/poorly worded questions · Student list export includes: name, email, standard, joined date, last active, total attempts